



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



SILVER OAK UNIVERSITY

Silver Oak College of Computer Application

Course: Foundation of Mathematics | Course Code: MDC5001C | Semester: 1

List of Tutorials

**UNIT 1: INTRODUCTION TO COMBINATORICS AND ITS
APPLICATIONS (CO Mapping: CO1)**

Tutorial Summary Table

Tutorial No.	Tutorial Title	Topics Covered	No. of Qs	CO
T1	Counting Fundamentals — Addition & Multiplication Rules	Basic Counting Principles: Addition and Multiplication Rules	7	CO1
T2	Arranging Objects — Permutations With & Without Repetition	Permutations (With and Without Repetition)	8	CO1
T3	Selecting Objects — Combinations & Related Identities	Combinations and Related Identities	8	CO1
T4	Expanding Powers — Binomial Theorem & Pascal's Triangle	Binomial Theorem and Pascal's Triangle	7	CO1
T5	Guaranteed Repeats — Pigeonhole Principle (Basic Form)	Pigeonhole Principle (Basic)	6	CO1
T6	Extending the Pigeonhole — Generalized Pigeonhole Principle	Pigeonhole Principle (Generalized)	6	CO1
T7	Counting the Complement — Inclusion-Exclusion Principle	Inclusion-Exclusion Principle	7	CO1
T8	No Fixed Points — Derangements & Applications	Derangements and Applications	7	CO1
T9	Counting Repeats — Multisets & Counting with Repetition	Counting with Repetition and Multisets	7	CO1
T10	Breaking Numbers Down — Integer Partitions & Compositions	Integer Partitions and Compositions	7	CO1

Tutorial No.	Tutorial Title	Topics Covered	No. of Qs	CO
T11	Solving Recurrences — Setting Up & Solving Recurrence Relations	Recurrence Relations and Their Solutions	7	CO1
T12	Encoding Sequences — Generating Functions & Applications	Generating Functions and Applications	7	CO1

Tutorial 1: Counting Fundamentals — Addition & Multiplication Rules

Topics covered: Basic Counting Principles — Addition and Multiplication Rules

Q.No	Question	Difficulty	Bloom's Level	CO
1	State the Addition Rule of counting, with a simple example.	Easy	Remember	CO1
2	State the Multiplication Rule of counting, with a simple example.	Easy	Understand	CO1
3	A restaurant menu has 4 starters, 6 mains, and 3 desserts. In how many ways can a customer choose one item from each category?	Moderate	Apply	CO1
4	A student can choose one elective from Mathematics (5 options) or CS (4 options), but not both. How many choices does the student have?	Moderate	Apply	CO1
5	How many 4-digit numbers can be formed using digits 0–9 if repetition is allowed and the first digit cannot be 0?	Moderate	Analyze	CO1
6	15 students study Python, 12 study Java, and 5 study both. How many students study at least one language?	Challenging	Analyze	CO1
7	A committee selects either a chairperson (5 candidates) or a co-chair (3 candidates), not both. In how many ways can this selection be made?	Challenging	Evaluate	CO1

Tutorial 2: Arranging Objects — Permutations With & Without Repetition

Topics covered: Permutations (With and Without Repetition)

Q.No	Question	Difficulty	Bloom's Level	CO
8	Define permutation and give an example distinguishing it from combination.	Easy	Remember	CO1
9	State the formula for the number of permutations of n distinct objects taken r at a time.	Easy	Understand	CO1
10	In how many ways can 6 different books be arranged on a shelf?	Moderate	Apply	CO1
11	How many distinct arrangements can be made from the letters of "COMPUTER"?	Moderate	Apply	CO1

Q.No	Question	Difficulty	Bloom's Level	CO
12	How many 3-letter codes can be formed from A–Z if repetition is allowed?	Moderate	Analyze	CO1
13	In how many ways can 5 people be seated in a row if two particular people must always sit together?	Challenging	Analyze	CO1
14	Find the number of ways to arrange the letters of "STATISTICS."	Challenging	Evaluate	CO1
15	A relay race has 4 runners chosen from a squad of 9. In how many ways can the runners be selected and ordered?	Challenging	Evaluate	CO1

Tutorial 3: Selecting Objects — Combinations & Related Identities

Topics covered: Combinations and Related Identities

Q.No	Question	Difficulty	Bloom's Level	CO
16	Define combination and state the formula for $C(n, r)$.	Easy	Remember	CO1
17	Explain the difference between a permutation and a combination, with an example.	Easy	Understand	CO1
18	In how many ways can a committee of 4 be selected from 10 people?	Moderate	Apply	CO1
19	From 7 men and 5 women, how many ways can a team of 3 men and 2 women be formed?	Moderate	Apply	CO1
20	Prove that $C(n, r) = C(n, n-r)$ using a combinatorial argument.	Moderate	Analyze	CO1
21	Verify Pascal's identity $C(n, r) = C(n-1, r-1) + C(n-1, r)$ for $n = 6, r = 3$.	Challenging	Analyze	CO1
22	A deck of 52 cards is used. How many 5-card hands contain exactly 2 aces?	Challenging	Evaluate	CO1
23	Using Vandermonde's identity, expand $C(m+n, r)$ in terms of $C(m, k)$ and $C(n, r-k)$, and verify for $m=3, n=2, r=2$.	Challenging	Create	CO1

Tutorial 4: Expanding Powers — Binomial Theorem & Pascal's Triangle

Topics covered: Binomial Theorem and Pascal's Triangle

Q.No	Question	Difficulty	Bloom's Level	CO
24	State the Binomial Theorem for $(x + y)^n$.	Easy	Remember	CO1
25	Construct the first 6 rows of Pascal's Triangle.	Easy	Understand	CO1
26	Expand $(x + y)^5$ using the Binomial Theorem.	Moderate	Apply	CO1

Q.No	Question	Difficulty	Bloom's Level	CO
27	Find the coefficient of x^4 in the expansion of $(2x - 3)^7$.	Moderate	Apply	CO1
28	Find the middle term in the expansion of $(x + 1/x)^8$.	Moderate	Analyze	CO1
29	Prove that the sum of coefficients in $(1 + x)^n$ equals 2^n .	Challenging	Analyze	CO1
30	Find the term independent of x in the expansion of $(x^2 - 2/x)^9$.	Challenging	Evaluate	CO1

Tutorial 5: Guaranteed Repeats — Pigeonhole Principle (Basic Form)

Topics covered: Pigeonhole Principle (Basic)

Q.No	Question	Difficulty	Bloom's Level	CO
31	State the Pigeonhole Principle in its basic form.	Easy	Remember	CO1
32	Show that if 13 people are in a room, at least 2 must share a birth month.	Easy	Understand	CO1
33	Prove that among any 5 integers, at least two have the same remainder when divided by 4.	Moderate	Apply	CO1
34	A drawer contains socks of 4 colors. How many socks must be pulled out blindly to guarantee a matching pair?	Moderate	Analyze	CO1
35	Show that among any 8 people, at least 2 were born on the same day of the week.	Challenging	Analyze	CO1
36	Prove that in any group of 27 English words, at least two begin with the same letter.	Challenging	Evaluate	CO1

Tutorial 6: Extending the Pigeonhole — Generalized Pigeonhole Principle

Topics covered: Pigeonhole Principle (Generalized)

Q.No	Question	Difficulty	Bloom's Level	CO
37	State the Generalized Pigeonhole Principle.	Easy	Remember	CO1
38	How many students must be in a class to guarantee that at least 5 share the same birth month?	Moderate	Understand	CO1
39	Prove that among any 100 integers, at least 10 have the same remainder modulo 9.	Moderate	Apply	CO1
40	A box contains balls of 6 colors. How many balls must be drawn to guarantee at least 4 of the same color?	Moderate	Analyze	CO1
41	Show that in any set of 51 integers between 1 and 100, some two must sum to 101.	Challenging	Analyze	CO1

Q.No	Question	Difficulty	Bloom's Level	CO
42	Using the Generalized Pigeonhole Principle, prove any sequence of n^2+1 distinct reals contains an increasing or decreasing subsequence of length $n+1$.	Challenging	Evaluate	CO1

Tutorial 7: Counting the Complement — Inclusion-Exclusion Principle

Topics covered: Inclusion-Exclusion Principle

Q.No	Question	Difficulty	Bloom's Level	CO
43	State the Inclusion-Exclusion Principle for two sets A and B.	Easy	Remember	CO1
44	State the Inclusion-Exclusion Principle for three sets A, B, and C.	Easy	Understand	CO1
45	In a survey of 100 students, 60 like tea, 50 like coffee, and 30 like both. How many like neither?	Moderate	Apply	CO1
46	How many integers from 1 to 500 are divisible by 3 or 5?	Moderate	Apply	CO1
47	A class of 40 students has 18 playing cricket, 20 playing football, and 10 playing both. How many play neither sport?	Moderate	Analyze	CO1
48	Given $ A =30$, $ B =25$, $ C =20$, $ A \cap B =10$, $ B \cap C =8$, $ A \cap C =7$, $ A \cap B \cap C =3$, find $ A \cup B \cup C $.	Challenging	Evaluate	CO1
49	Find the number of surjective (onto) functions from a 4-element set to a 3-element set using Inclusion-Exclusion.	Challenging	Create	CO1

Tutorial 8: No Fixed Points — Derangements & Applications

Topics covered: Derangements and Applications

Q.No	Question	Difficulty	Bloom's Level	CO
50	Define a derangement, with a suitable example.	Easy	Remember	CO1
51	Compute $D(4)$ by listing all valid derangements of $\{1,2,3,4\}$.	Easy	Understand	CO1
52	Find the number of derangements of 5 objects, $D(5)$, using the recurrence formula.	Moderate	Apply	CO1
53	In how many ways can 6 letters be placed into 6 addressed envelopes so that no letter goes into its correct envelope?	Moderate	Apply	CO1
54	Derive the formula for $D(n)$ using the Inclusion-Exclusion Principle.	Challenging	Analyze	CO1
55	Compute the probability that a random permutation of 6 elements is a derangement.	Challenging	Evaluate	CO1

Q.No	Question	Difficulty	Bloom's Level	CO
56	Compare $D(n)$ with $n!$ for large n , and explain the $\approx 1/e$ significance in real-life scheduling/allocation problems.	Challenging	Evaluate	CO1

Tutorial 9: Counting Repeats — Multisets & Counting with Repetition

Topics covered: Counting with Repetition and Multisets

Q.No	Question	Difficulty	Bloom's Level	CO
57	State the formula for the number of ways to distribute n identical objects into r distinct groups.	Easy	Remember	CO1
58	How many ways can 10 identical candies be distributed among 4 children?	Easy	Understand	CO1
59	Find the number of non-negative integer solutions to $x_1 + x_2 + x_3 = 15$.	Moderate	Apply	CO1
60	How many ways can 5 identical balls be placed into 3 distinct boxes such that no box is empty?	Moderate	Apply	CO1
61	How many distinct 4-letter "words" can be formed from the multiset $\{A, A, B, B, C\}$?	Moderate	Analyze	CO1
62	A bakery has 6 types of pastries. How many ways can a customer select a box of 10 pastries (repetition allowed)?	Challenging	Analyze	CO1
63	Find the number of solutions to $x+y+z+w = 20$ where $x \geq 2$, $y \geq 0$, $z \geq 3$, $w \geq 0$.	Challenging	Evaluate	CO1

Tutorial 10: Breaking Numbers Down — Integer Partitions & Compositions

Topics covered: Integer Partitions and Compositions

Q.No	Question	Difficulty	Bloom's Level	CO
64	Define an integer partition, with an example.	Easy	Remember	CO1
65	Define an integer composition and distinguish it from a partition, with an example for $n = 4$.	Easy	Understand	CO1
66	List all partitions of the integer 6.	Moderate	Apply	CO1
67	Find the number of compositions of 5 (ordered sums of positive integers).	Moderate	Apply	CO1
68	How many ways can 8 be partitioned into exactly 3 parts?	Moderate	Analyze	CO1
69	Find the number of partitions of 10 into distinct parts.	Challenging	Analyze	CO1
70	Using generating functions or direct enumeration, find $p(7)$, the number of partitions of 7.	Challenging	Evaluate	CO1

Tutorial 11: Solving Recurrences — Setting Up & Solving Recurrence Relations

Topics covered: Recurrence Relations and Their Solutions

Q.No	Question	Difficulty	Bloom's Level	CO
71	Define a recurrence relation, with an example.	Easy	Remember	CO1
72	Solve the recurrence relation $a_n = 2a_{n-1} + 1$, with $a_0 = 1$.	Easy	Understand	CO1
73	Find the closed-form solution of the Fibonacci recurrence $F_n = F_{n-1} + F_{n-2}$, with $F_0=0$, $F_1=1$.	Moderate	Apply	CO1
74	Solve $a_n - 5a_{n-1} + 6a_{n-2} = 0$ given $a_0=2$, $a_1=5$ (homogeneous, distinct roots).	Moderate	Apply	CO1
75	Set up a recurrence relation for the number of ways to climb n stairs taking 1 or 2 steps at a time, and solve it.	Moderate	Analyze	CO1
76	Solve the non-homogeneous recurrence $a_n = 3a_{n-1} + 2^n$.	Challenging	Analyze	CO1
77	Find the recurrence relation and initial conditions for the Tower of Hanoi problem with n disks, and solve it.	Challenging	Evaluate	CO1

Tutorial 12: Encoding Sequences — Generating Functions & Applications

Topics covered: Generating Functions and Applications

Q.No	Question	Difficulty	Bloom's Level	CO
78	Define the generating function of a sequence, with an example.	Easy	Remember	CO1
79	Find the generating function for the sequence $a_n = 3$ for all $n \geq 0$.	Easy	Understand	CO1
80	Use generating functions to solve the recurrence $a_n = a_{n-1} + 2$, $a_0 = 1$.	Moderate	Apply	CO1
81	Find the generating function for the number of ways to make change for n rupees using coins of denomination 1, 2, and 5.	Moderate	Apply	CO1
82	Derive the generating function for the Fibonacci sequence and use it to find a closed-form expression for F_n .	Challenging	Analyze	CO1
83	Use generating functions to find the number of solutions to $x_1 + x_2 + x_3 = 10$ where each $x_i \leq 4$.	Challenging	Evaluate	CO1
84	Find the coefficient of x^n in the expansion of $1/(1-x)^3$, and interpret its combinatorial meaning.	Challenging	Create	CO1

